Nikos Gkanatsios

Next-Generation Embodied Intelligence

Generative Decision Intelligence • Robot Learning • Multimodal Perception

CMU Robotics PhD | Autonomy Research & Production Systems (Wayve • Tesla • NVIDIA)

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Industry Research Experience

Research contributions spanning model-based reinforcement learning, scalable policy learning, post-training optimization and generative modeling for autonomous systems.

Wayve — Applied Scientist — Apr 2026 - Present

- Researching model-based reinforcement learning and world models for autonomous driving.
- Contributing to next-generation foundation models for embodied intelligence.

Tesla — Sr Autopilot Machine Learning Engineer — Sep 2025 - Feb 2026

- Developed reinforcement learning-based closed-loop driving policies using large-scale fleet data.
- Contributed to post-training adaptation methods for improving real-world driving behavior.
- Designed data selection strategies supporting scalable policy learning.
- Worked within production-scale training and evaluation pipelines for deployment systems.

NVIDIA — Research Intern — June 2024 - May 2025

- Developed flow-based generative methods for 3D manipulation policy learning.
- Enabled scaling of 3D policy learning in model capacity and compute efficiency through optimized distributed training and dataset management pipelines.

Deeplab — Machine Learning Engineer — July 2018 - July 2020

- Worked on visual-semantic scene understanding using graph-based perception modeling.

METIS Cybertechnology — Artificial Intelligence Engineer — March 2017 - July 2018

- Developed and deployed production machine learning models with end-to-end pipeline ownership.

Research Themes and Representative Publications

Research contributions spanning visual-language perception, generative decision modeling and scalable policy learning for embodied intelligence systems. Full publication list available on my Google Scholar profile.

Multimodal Embodied Perception

Learning spatial-aware vision-language representations for grounded scene understanding.

- **BUTD-DETR** — Vision-language grounding in images and point clouds — *ECCV 2022*
- **ODIN** — Unified 2D–3D segmentation for spatial scene understanding — *CVPR 2024*

Generative Autonomy Learning

Converting perception models into generative policy and planning models for decision making.

- **3D Diffuser Actor** — Policy diffusion with 3D scene representations — *CoRL 2024*
- **Diffusion-ES** — Gradient-free diffusion planning for autonomous driving — *CVPR 2024*
- **ChainedDiffuser** — Generative hierarchical planning for long-horizon robotic tasks — *CoRL 2023*
- **EBM Planner** — Energy-based compositional goal generation for manipulation planning — *RSS 2023*
- **3D FlowMatch Actor** — Flow-matching policies for scalable 3D manipulation — *IROS 2026*

Geometry-Aware Autonomy Learning

Leveraging spatial structure for efficient policy learning and decision optimization.

- **Act3D** — Geometry-aware representations for language-conditioned manipulation *CoRL 2023*
- **3D Diffuser Actor** — Spatially-aware generative policies *CoRL 2024*

Education

Carnegie Mellon University Pittsburgh, PA, USA
PhD, Robotics Institute 2020 – 2025
Advisor: Katerina Fragkiadaki
Thesis: Unified 3D Perception and Generative Control for Generalist Robots

National Technical University of Athens (NTUA) Athens, Greece
Diploma, Electrical & Computer Engineering 2011 – 2017
Advisor: Petros Maragos
GPA: 9.22 / 10, ranked 1st in Electronics & Systems major

Learning and Embodied Intelligence Capabilities

Autonomy Decision Learning
Scalable reinforcement learning, imitation learning and planning for embodied agents.

Generative Policy and Planning Models
Generative modeling for policy learning and goal-conditioned decision making using diffusion, flow matching and energy-based models.

Embodied Multimodal Perception
Vision-language grounding, RGB-D perception and 3D scene representation learning for embodied intelligence systems.

Large-Scale Learning Systems
Distributed training, mixed precision optimization, large-scale dataset curation and high-throughput ML training pipelines using PyTorch.

Peer Review

Reviewer for leading venues across machine learning, computer vision and robotics:
Machine Learning: NeurIPS, ICML, ICLR, AAAI
Computer Vision: CVPR, ICCV, ECCV, ACCV, BMVC
Robotics: RSS, CoRL, ICRA, IROS
Journals: TPAMI, RA-L